K L (Deemed to be) University

DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY



WORD-LEVEL PLAGIARISM CHECKER FOR SHORT DOCUMENTS

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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DECLARATION

The Mini Project Report entitled “Word-Level Plagiarism Checker for Short Documents” is a record of bonafide work carried out by Hitesh Suthar (2520090078) and Vasudeva (2520090136) in partial fulfillment of the requirements for the course Data Structures and Algorithms -3 (25CS2103E). The project is based on a word-level approach for detecting possible plagiarism in short documents.

The project documentation has been prepared from the supplied project reference presentation. No claim is made here about test results or implementation details that are not present in that reference.

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CERTIFICATE

This is to certify that the mini project based report entitled “Word-Level Plagiarism Checker for Short Documents” is a bonafide work done and submitted by Hitesh Suthar (2520090078) and Vasudeva (2520090136) in partial fulfillment of the requirements for the course Data Structures and Algorithms -3 (25CS2103E), during the academic year 2026-2027, under the guidance of Chandusha Kanda.

Signature of the Supervisor

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Signature of the HOD

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Signature of the External Examiner

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ACKNOWLEDGEMENT

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We also thank the faculty members and staff of K L (Deemed to be) University for providing the academic environment and facilities required to complete the project.

Finally, we thank our classmates, friends and family for their encouragement and cooperation throughout the project work.

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1. ABSTRACT

The Word-Level Plagiarism Checker for Short Documents is a software system designed to identify copied or duplicated content in short documents. The system compares the words and phrases of a submitted document with reference or existing text. It first extracts the text and preprocesses it by standardizing the content, removing unnecessary characters, and preparing it for comparison. Tokenization divides the document into individual words or meaningful units. The system then performs word-level comparison to identify matching words, repeated words, and common sequences. A similarity percentage is calculated from the matching content and is used as an indicator of possible plagiarism. The system can present matched content, the similarity score, and a plagiarism indication in a report, helping users reduce the effort required for manual checking. The reference presentation specifies Java as the programming language and Java-based text processing as the implementation environment. The approach is intended for assignments, reports, articles, and other short documents. Future extensions may incorporate Natural Language Processing and Machine Learning to detect deeper forms of similarity.

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2. INTRODUCTION

Plagiarism detection is the process of identifying content in a document that matches or closely resembles content from another source. For short documents, a lightweight word-level approach can provide a simple way to highlight possible duplication without requiring a complex detection pipeline.

The proposed Word-Level Plagiarism Checker compares a submitted document against reference text. The workflow begins with text extraction and preprocessing. Preprocessing includes lowercase conversion, punctuation removal, stop-word removal, and tokenization. The resulting tokens are compared to identify matching words and consecutive word sequences.

After comparison, the system calculates a similarity percentage and analyzes the score to determine whether the document shows possible plagiarism. A report is then generated containing the matched content, the similarity score, and a plagiarism indication. This makes the output easier for a user to understand.

The project is focused specifically on short documents. The reference presentation identifies NLP and Machine Learning as possible future extensions for detecting deeper similarities beyond direct word-level matches.

3. LITERATURE SURVEY

The supplied project presentation does not name individual plagiarism-checking products, research papers, or website URLs. Therefore, the literature survey below summarizes the technical approaches explicitly identified in the reference material rather than inventing external citations.

Text Preprocessing: Text preprocessing prepares documents for comparison by converting text to lowercase, removing punctuation and stop words, and tokenizing the remaining content. This creates a more standardized representation for matching.

Word Matching: Word matching compares individual words between the submitted and reference documents. Matching words provide a direct indication of shared content.

Sequence Matching: Sequence matching considers common words and consecutive word sequences. It can identify repeated phrases in addition to isolated matching words.

Similarity Measurement: Similarity measurement uses the amount of matching content to calculate a similarity percentage. The percentage serves as a compact indicator of how much content overlaps.

Plagiarism Reporting: A plagiarism report presents the matched content, similarity score, and possible plagiarism indication so that the user can review the result.

Advanced Methods: The reference material identifies Natural Language Processing and Machine Learning as future methods that could be used to detect deeper or less direct similarities.

Research gap addressed by this project: the project emphasizes a simple, understandable word-level workflow for short documents, reducing the complexity of manual checking while leaving advanced semantic detection for future work.

4. PROBLEM STATEMENT AND OBJECTIVES

Problem Statement: Manual identification of copied words and phrases in short documents can be time-consuming; a simple automated method is needed to compare text and indicate possible plagiarism.

Problem Description: The system must process a submitted document and reference text, identify matching words or sequences, calculate a similarity percentage, and present the result as an understandable plagiarism report.

Objectives

1. Accept a short document and reference documents as input.
2. Extract text from the supplied documents.
3. Preprocess text using lowercase conversion, punctuation removal, stop-word removal, and tokenization.
4. Compare words and common sequences between the submitted and reference text.
5. Calculate a similarity percentage from the matching content.
6. Analyze the similarity score to indicate possible plagiarism.
7. Generate a report containing matched content and the similarity result.
8. Reduce the manual effort involved in checking short documents.

5. HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Requirements

- Computer or laptop.
- Basic processor.
- Minimum 4 GB RAM.
- Sufficient storage.
- Keyboard, mouse, and monitor.

Software Requirements

- Operating System: Windows.
- Programming Language: Java.
- Development Environment: Java IDE.
- Text Processing: Java-based text processing.
- Input: Short text documents.
- Output: Similarity percentage and plagiarism report.

These requirements are taken directly from the supplied project presentation. The presentation does not specify a minimum processor model, exact storage capacity, or a particular Java IDE, so those details are intentionally not added.

6. IMPLEMENTATION

The implementation follows the eight-step methodology presented in the project reference. The modules below organize that methodology into functional components.

Module 1 – Input and Text Extraction
Accept the submitted document and reference documents, then extract their textual content for processing.

Module 2 – Text Preprocessing
Convert text to lowercase, remove punctuation and stop words, and tokenize the text into individual words or meaningful units.

Module 3 – Word-Level Comparison and Similarity
Compare tokens to identify matching words, repeated words, and common consecutive sequences, then calculate the similarity percentage.

Module 4 – Plagiarism Analysis and Report Generation
Analyze the similarity score, identify possible plagiarism, generate matched-content information and the score, and display the report.

Methodology Flow

Input Document → Text Extraction → Text Preprocessing → Word-Level Comparison → Similarity Calculation → Plagiarism Detection → Report Generation → Display Results

Implementation note: The reference presentation specifies Java and Java-based text processing but does not include source code. Accordingly, this report documents the implementation modules and workflow without claiming that a specific code listing is the exact code used by the project.

7. RESULTS & DISCUSSION

According to the project reference, the expected application output is a plagiarism report containing the detected matching content, a similarity percentage, and a plagiarism indication. The workflow is intended to provide users with a quick and understandable way to review possible duplication in short documents.

The reference material confirms the following functional outcomes:

- Matching words and phrases can be identified through word-level and sequence comparison.
- A similarity percentage is calculated from the matching content.
- The result can be presented as a plagiarism report.
- The approach reduces the effort required for manual checking.
- The design can be extended with NLP and Machine Learning for deeper similarity detection.

Result-data limitation: The supplied presentation contains no numerical test cases, measured accuracy values, sample similarity percentages, or screenshots of the running checker. Therefore, this report does not fabricate performance numbers or claim specific experimental results.

For the final submission, screenshots of the actual application run can be inserted in this section, with each screenshot accompanied by a short explanation of the input, detected matches, similarity score, and final report.

8. CONCLUSION & FUTURE SCOPE

The Word-Level Plagiarism Checker for Short Documents provides a simple method for identifying possible plagiarism in short text documents. It uses text extraction and preprocessing followed by word-level and sequence-based comparison. The matching content is converted into a similarity percentage and presented through an understandable plagiarism report. The approach can reduce the manual effort required to inspect short documents.

Future Scope

- Integrate Natural Language Processing (NLP) techniques to detect deeper similarities.
- Apply Machine Learning methods to improve detection of less-direct or semantically similar content.
- Extend the system beyond short documents to larger documents.
- Provide richer reports that help users review the exact source of matching content.
- Evaluate the system using a larger set of test documents and quantitative performance measures.

The first two future directions are explicitly identified in the reference presentation; the remaining items are reasonable extensions of the documented workflow and are presented as proposed future work rather than completed features.

9. REFERENCES

The supplied project presentation does not provide external URLs or named references. The following source is the basis used for this documentation:

1. Project reference presentation: “Word-Level Plagiarism Checker for Short Documents,” supplied by the project team.

2. Project report template: “DSA-3 Project Report (25CS2103E),” K L (Deemed to be) University, A.Y. 2026-2027.

Note for final submission: If the team used external websites, papers, libraries, or plagiarism-detection systems during development, their URLs/citations should be added here. They were not present in the supplied PPT, so no external references have been invented.